

Introduction

This report evaluates the investment potential of NDQ in combination with XNT from April 2021 to March 2026. The analysis focuses on NDQ's role as a growth-oriented diversification asset within an Australian equity portfolio and examines whether exposure to US technology companies can improve the portfolio's overall risk-return performance. Historical return analysis, rolling beta estimation, portfolio optimisation, and market trend analysis were conducted to assess NDQ's market behaviour, systematic risk, and portfolio contribution.

The results suggest that NDQ provides strong growth potential due to its exposure to major US technology companies including Nvidia, Apple, Microsoft, Amazon, and Meta. In addition, NDQ offers international diversification benefits through exposure to the US market and different interest rates and currency environments. However, the ETF also demonstrates higher volatility and greater sensitivity to technology sector movements compared with XNT. The report further considers the major opportunities and risks that may affect NDQ's future performance.

1.1 Monthly Return Chart Analysis

The monthly return data of NDQ and XNT indicate that both assets are volatile. Returns both vary over time, and the fluctuation range of NDQ is usually larger, indicating higher risk and higher returns compared to XNT.

1.2 Cumulative Return Curve Explanation

The cumulative return graph shows that the long-term growth rate and yield of NDQ are higher than those of XNT. However, the short-term growth of NDQ is not as good as that of XNT, indicative of NDQ's higher volatility.



2.1 Rolling Beta Estimation

- Estimation of the 24-month rolling beta for NDQ against XNT, from March 2023 to April 2026.
- Plotting the rolling beta



- Beta is the systematic risk of NDQ against the market proxy, XNT. From 2023 to 2026, we see that there are fluctuations in beta. The beta for this asset ranged from around 0.38 to -0.03. Tracking the 37 months of rolling beta, there was a general maintenance of a beta around the 0.3

range until June 2024, in which it started declining towards -0.03 in Jan 2025. The Beta then made an upwards re correction and has been slowly dropping since. The total 60-month beta for this sample was 0.53. Thus, it is shown our asset's systematic risk relative to XNT over time and is has less systematic risk relative to XNT as a market proxy. An inclusion of this asset will reduce the overall volatility of our investor's portfolio.

This is explained by the difference in composition of NDQ, which tracks the largest 100 non-financial companies on the NASDAQ stock exchange (Gleeson, 2024). This ETF provides investors overseas exposure and towards different sectors, mostly technology and growth stocks, which XNT lacks. XNT being Australian means that changes to domestic interest rates affect XNT significantly, whereas NDQ is largely driven by US fed policy, rather than the RBA's. NDQ's heavy US weighting and concentration in growth stocks means that US rate changes will have a large effect on the valuation of these companies, affecting NDQ disproportionately more (Nasdaq, 2022).

3.1 CAPM Analysis

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> summary(capm_model)

Call:
lm(formula = NDQ_excess ~ XNT_excess, data = capm_data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.062e-03 -3.008e-04 -2.872e-05  2.475e-04  1.255e-03

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0001153  0.0001220   -0.945   0.349
XNT_excess   0.9346949  0.0450215   20.761 <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0005008 on 58 degrees of freedom
Multiple R-squared:  0.8814,    Adjusted R-squared:  0.8794
F-statistic: 431 on 1 and 58 DF,  p-value: < 2.2e-16
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The CAPM regression results indicate that NDQ has a beta of approximately 0.935, which is close to 1. This suggests that NDQ's systematic risk is broadly comparable to the market portfolio, although its risk level is slightly lower than the market benchmark.

Furthermore, the estimated alpha value is close to zero, indicating that the performance of NDQ is mainly determined by market-related factors rather than abnormal excess returns. The regression analysis also yields an R^2 value of approximately 0.881, suggesting that a significant portion of the changes in NDQ's returns can be explained by broader market fluctuations.

Although NDQ remains highly connected to overall market conditions, it may still provide diversification benefits through its exposure to US technology and growth-oriented sectors that are underrepresented within the Australian equity market

3.2 Correlation & Sharpe ratio analysis

Correlation	Sharpe ratio
0.9388268	0.1647134

The correlation between the excess returns of NDQ and XNT is approximately 0.939, indicating a strong positive relationship between the two assets. Although NDQ remains highly influenced by broader market movements, it may still provide diversification benefits through its exposure to US technology and growth-oriented sectors that are underrepresented in the Australian market.

Furthermore, the Sharpe ratio of NDQ during the sample period was approximately 0.165, which was positive. This indicates that the exchange-traded fund provided investors with positive risk-

adjusted returns, although the value of the Sharpe ratio suggests that the returns were accompanied by a considerable degree of volatility.

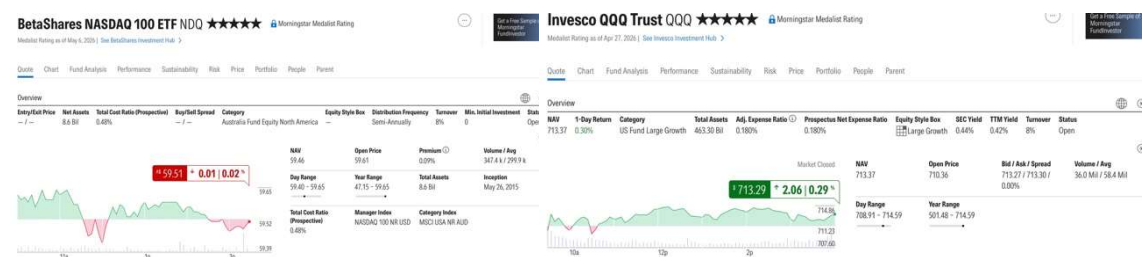
3.3 NAV Analysis

A Net Asset Value analysis was used because NDQ is an ETF, and the key indicator for ETF price efficiency is comparing market price with its NAV.

On May 12th, NDQ's market price was A\$59.51, while its NAV was A\$59.46, implying a premium of approximately 0.09% (Morningstar, 2026a). This small premium shows that NDQ closely tracks the value of its underlying holdings and trades efficiently.

NDQ was also compared against the largest US-listed Nasdaq 100 ETF, Invesco QQQ Trust. Both ETFs track the Nasdaq 100 index, so they provide very similar exposure to large U.S. technology companies like Apple, Microsoft, Nvidia, Amazon and Meta (Morningstar, 2026a; Morningstar, 2026b). QQQ traded at a small discount to NAV (-0.011%) and showed very strong pricing efficiency (Morningstar, 2026b). The difference between QQQ's deviation from NAV was marginally smaller than NDQ's but is economically meaningless since both ETFs trade very close to the value of their underlying portfolios.

However, NDQ may be more suitable for the client because it allows Australian investors to access major US technology companies in Australian dollars through a familiar domestic trading platform while maintaining pricing quality comparable to QQQ.



3.4 Upside & downside risks

NDQ provides diversification outside the Australian market by giving the portfolio exposure to large US technology companies, which may support stronger long-term growth. In addition, continued investment in artificial intelligence and digital infrastructure may support future earnings growth.

However, NDQ is heavily weighted towards large United state technology companies, so weaker performance in the tech sector could materially impact returns. Additionally, many Nasdaq 100 companies trade at relatively high valuations, creating valuation risk if earnings growth slows or interest rates rise. These risks can be monitored through company earnings performance and valuation ratios.

3.5 Recommendation

Based on the CAPM, correlation, Sharpe ratio, and NAV analysis, NDQ appears to be a suitable addition to the client's Australia-focused portfolio. The NDQ provides exposure to major US technology and growth companies that are underrepresented in the Australian market, while also offering geographic diversification and positive long-term growth potential.

The CAPM results suggest that NDQ's systematic risk is broadly comparable to the market, while the positive Sharpe ratio indicates positive risk-adjusted returns during the sample period. In addition, the NAV analysis demonstrates that NDQ trades very closely to the value of its underlying holdings, indicating strong pricing efficiency.

Although certain risks remain, particularly the concentration in large US technology companies and high growth stock valuations, NDQ still appears to be an appropriate long-term investment option for the client's portfolio.

Conclusion

Overall, the analysis demonstrates that NDQ can enhance the portfolio's growth potential and improve international diversification when combined with Australian equities. The ETF provides exposure to leading global technology companies and has delivered strong long-term return performance throughout much of the analysis period. In addition, NDQ reduces concentration in the Australian market by increasing exposure to the US technology sector and different macroeconomic conditions.

Although risks such as higher US interest rates, technology sector volatility, and exchange rate fluctuations may affect future performance, NDQ continues to offer meaningful diversification and long-term growth opportunities for investors. Given the continued expansion of the global technology industry and the strong market position of its major holdings, NDQ represents a suitable investment option for investors seeking higher growth and broader portfolio diversification.

References list

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